Post-COVID Lung Disease in Children

19

Introduction

SARS-CoV-2 infection in children is primarily mild compared to the adult population and usually resolves in 4 weeks. Some patients may experience persistent symptoms attributed to COVID infection beyond 12 weeks and cannot be explained by alternative diagnosis.

It is known by different names:

- ☑ Post COVID-19 syndrome
- ☑ Long COVID, long-haul COVID-19
- ☑ Chronic COVID-19
- ☑ Post-acute sequelae of SARS-CoV-2 (PASC), etc.

This condition is well described in adults; information in children is scarce. Post-COVID lung disease (PCLD) is a spectrum of post-COVID-19 syndrome with predominant respiratory manifestation. After an extensive search in published literature, the following information is compiled. The recommendations are based on available current evidence.

Prevalence

- ☑ The prevalence of post-COVID-19 syndrome in adults is variable and ranges from 5 to 80%.
- ☑ There is very limited information on the prevalence of long COVID in children. Based on a meta-analysis (including 17 studies in children and young people with COVID) reported persistent cough and dyspnea in 17% (95% CI 7–27) and 43% (95% CI 18–68), respectively.
- ☑ Another review of 14 studies of long COVID in children and adolescents reported persistent symptoms of running/congested nose, chest pain/tightness and sleep disturbances in 1–12%, 1–31%, and 2–63%, respectively.
- ☑ The other clinical manifestations included: fever (18%), fatigue (47%), headache (35%), cognitive dysfunction (26%), myalgia (25%), pain abdomen (25%), diarrhea (15%) and loss of smell (18%).
- ☑ However, a comparison of prevalence of symptoms in COVID infected and noninfected populations showed no difference, suggesting that some of the long-COVID symptoms may be an indirect effect of pandemic (school closure, lock-down, etc.).

Pathophysiology

- ☑ The pathophysiology of PCLD is not precisely known. PCLD can occur due to direct damage to the respiratory system or other system involvement (e.g., cardiac, rheumatological, nutritional, etc.).
- ☑ The postulated mechanisms are lung damage due to initial infection, ongoing virus-host interaction, persistent hyperinflammation, poor antibody response, and an exaggerated immune response leading to autoimmunity.

Risk Factors

There is little known about risk factors for PCLD in children. The reported factors associated with persistent symptoms in children are:

- ☑ Increasing age (6–18 years),
- ☑ Female gender,
- ☑ Worse pre-infection health,
- ☑ Presence of allergic diseases,
- ☑ Severe infection, and
- ☑ Prolonged hospitalization.

Types and Severity

The exact long-term effect of SARS- CoV2 in children is still unknown. The reported and possible conditions are:

- ☑ Reactive airway disease;
- ☑ Interstitial lung disease (ILD);
- ☑ Postinfectious bronchiolitis obliterans (PIBO);
- ☑ Diffuse alveolar hemorrhage; and
- ☑ Exercise intolerance.

Diagnosis

No laboratory test can distinguish PCLD from other etiology. On the other hand, an excessive investigation may be uninformative, costly, and harm the children. Therefore, one should adopt a conservative approach with the minimum investigation and focus on optimizing health conditions. Assessment of post-COVID lung disease is given in **Table 1**.

TABLE 1: Assessment of post-COVID lung disease.

Sl. No	*Assessment*	*Details*
1.	Clinical history and examination	Look for dyspnea, cough, chest pain, exercise intolerance, fatigue, sore throat, nasal congestion, voice change, etc.
2.	Chest imaging	Warranted in persistent symptoms without clinical improvement or with the development of new symptoms
3.	Pulmonary function test with bronchodilator reversibility (BDR)	May consider in children > 5 years. It will help in knowing the pattern of lung involvement, i.e., obstructive vs restrictive Positive BDR will indicate reactive airway disease
4.	Exercise test: 6-minute walk test (6MWT) and cardiopulmonary exercise testing (CPET)	In one study, impaired 6 MWT was observed in 66.6% of children (n = 9) with long COVID Other conditions like heart disease and thromboembolic disease should be ruled out before CPET
5.	Plethysmography and diffusing study (DLCO)	These can help diagnose restrictive lung disease as indicated by air trapping (RV/TLC > 30%)

Treatment

- ☑ There is no study on post-COVID lung disease treatment in children. The treatment is symptomatic and supportive as per the disease pattern. For airway obstruction, bronchodilators with or without inhaled steroids may be useful.
- ☑ Children of documented interstitial lung disease (ILD) is similar to non-COVID ILD, including systemic steroids and/or other immunosuppressants/immunomodulator drugs with monitoring.

Prevention

The information on preventive measures for post-COVID syndrome/post-COVID lung disease is scarce. Most of the risk factors (as mentioned above) for long-COVID in children are non-modifiable except pre-infection health. Therefore, children with any existing disease should be managed appropriately and aggressively to keep the disease under control.

Prognosis

There is a lack of studies on long-term follow-up of children with COVID, and little is known how long the effects of COVID may last. A register-based study from Norway reported an increase in healthcare use up to 6 months in 1–5 years aged children for respiratory and general conditions.

Role of Vaccination

Information on the impact of vaccination on post-COVID sequelae is lacking. A preprint study in adults reported that vaccination was associated with decreased acute COVID complications and long-term sequelae. It was more evident in adults < 60 years of age rather than > 60 years of age.

Future Direction

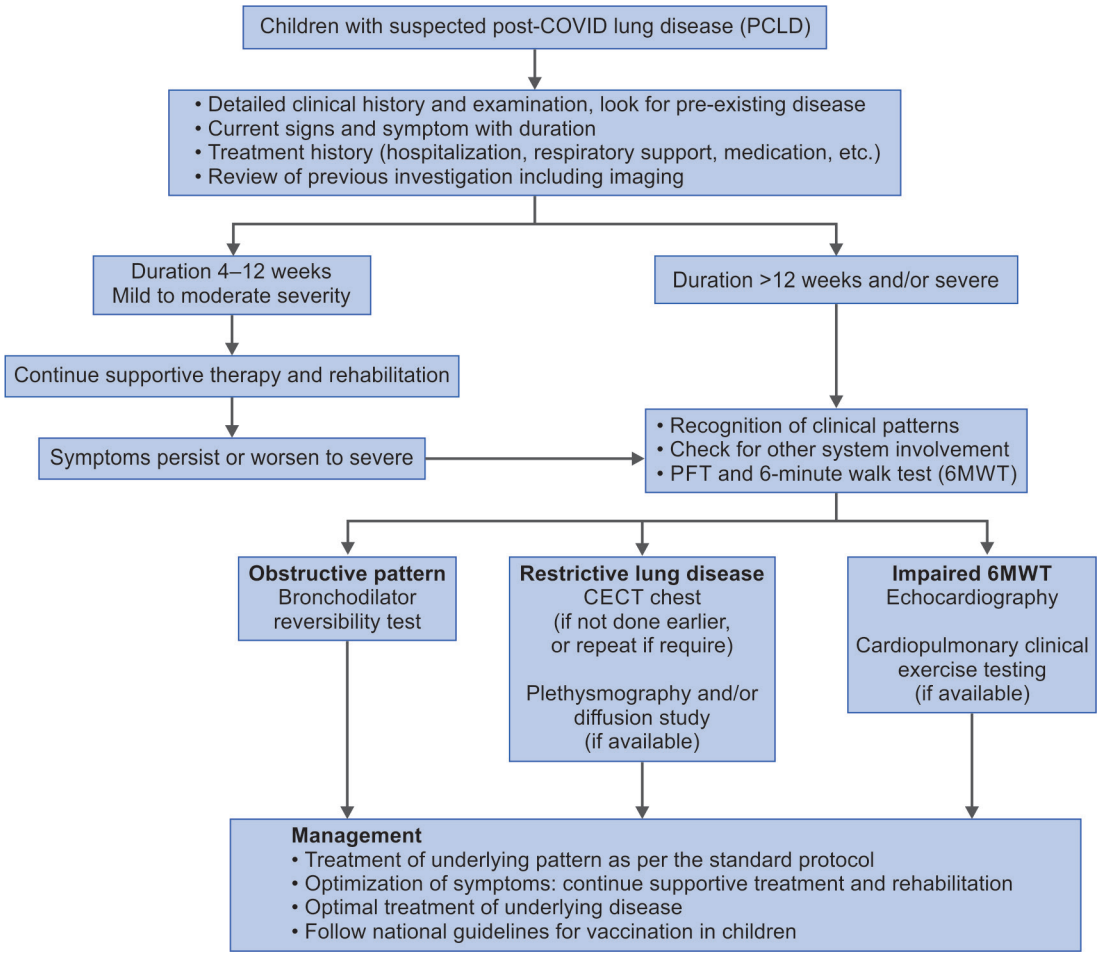
- ☑ As COVID-19 is a newly emerging illness, its long-term sequelae needs evaluation, including pathophysiology, effect on overall health, specifically effect on lung health and impact of vaccination on long-term sequelae.
- ☑ Multiple centers should develop cohort of children with COVID-19 infection of different severity and prepare uniform meticulous follow-up plans so that data can be compiled later on to obtain meaningful results.

Lung Involvement in Multisystem Inflammatory Syndrome in Children (MIS-C)

MIS-C is a newly defined illness that develops 2–4 weeks after COVID-19, and it can affect the lungs. Due to limited data, it is difficult to define lung lesions in MIS-C. Findings that may be in MIS-C are nonspecific and include consolidation, cavitary lesions, pleural effusion, ground-glass opacities on imaging etc.

Approach to PCLD

Flowchart 1: Proposed flow diagram for children with suspected post-COVID lung disease.



Conclusion

- ☑ COVID-19 may cause the post-COVID-19 syndrome, including post-COVID lung disease.
- ☑ It is well defined in adults, but there are limited studies in children with variable definitions, inclusion criteria and follow-up periods.
- ☑ There is no information on treatment and preventive measures for long COVID in children.
- ☑ Till more studies become available, optimal treatment of underlying respiratory disease, supportive/symptomatic treatment for long COVID and follow national guidelines regarding vaccination of children.
- ☑ Based on the current level of evidence, an algorithmic approach for children with suspected post-COVID lung disease is given in **Flowchart 1**.
- ☑ It is expected that with new information, we will better understand pathophysiology, clinical manifestations, treatment, and preventive aspects of long COVID in children.

Further Reading

- ☑ Behnood SA, Shafran R, Bennett SD, Zhang AXD, O'Mahoney LL, Stephenson TJ, et al. Persistent symptoms following SARS-CoV-2 infection amongst children and young people: a meta-analysis of controlled and uncontrolled studies. J Infect. 2021:S0163-4453(21)00555-7.
- ☑ Post-COVID-19 Conditions in Children and Adolescents. COVID-19 interim guideline. American Academy Pediatrics (AAP). Accessed from: https://www.aap.org/en/pages/2019-novel-coronavirus-covid-19-infections/clinical-guidance/post-covid-19-conditions-in-children-and-adolescents/
- ☑ Rostad BS, Shah JH, Rostad CA, Jaggi P, Richer EJ, Linam LE, et al. Chest radiograph features of multisystem inflammatory syndrome in children (MIS-C) compared to pediatric COVID-19. Pediatr Radiol. 2021;51:231-8.
- ☑ Taquet M, Dercon Q, Harrison PJ. Six-month sequelae of post-vaccination SARS-CoV-2 infection: a retrospective cohort study of 10,024 breakthrough infections. MedRxiv. 2021:2021.10.26.21265508.
- ☑ Zimmermann P, Pittet LF, Curtis N. How common is long COVID in children and adolescents? Pediatr Infect Dis J. 2021;40:e482-7.

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Acute Watery Diarrhea

Lead Author

Nirmala Cherukuri

Co-Authors

Nishanth Wadhwa, Prasanth K Sobhan

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

© Indian Academy of Pediatrics

IAP Standard Treatment Guidelines Committee

Chairperson

Remesh Kumar R

IAP Coordinator

Vineet Saxena

National Coordinators

SS Kamath, Vinod H Ratageri

Member Secretaries

Krishna Mohan R, Vishnu Mohan PT

Members

Santanu Deb, Surender Singh Bisht, Prashant Kariya, Narmada Ashok, Pawan Kalyan